

Design and Simulation of Microneedle using COMSOL Multiphysics Software for Transdermal Drug Delivery

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Abstract: Microneedle technology has emerged as a promising minimally invasive approach for transdermal drug delivery, overcoming the limitations of conventional hypodermic needles and oral drug administration. This paper presents a comprehensive computational study of microneedle arrays using COMSOL Multiphysics, focusing on geometric design optimization, mechanical behaviour during skin insertion, and drug diffusion mechanisms. A multilayer skin model incorporating stratum corneum, epidermis, and dermis with distinct material properties is developed to simulate realistic insertion mechanics and mass transport. Finite element analysis is employed to evaluate stress distribution, deformation patterns, and drug concentration profiles for different microneedle geometries including conical, pyramidal, and hollow designs. Parametric studies are conducted to optimize key parameters such as needle height (300-1000 μm), tip radius (1-25 μm), base diameter, and array density. The coupled multi physics simulations demonstrate that conical microneedles with optimized tip geometry achieve optimal stress distribution and drug delivery efficiency. The results provide valuable insights for the design of safe, effective, and patient-friendly microneedle-based drug delivery systems.

Keywords: Microneedle, Transdermal Drug Delivery, COMSOL Multiphysics, Finite Element Analysis, Drug Diffusion, Skin Modelling, Design Optimization



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